Rebuttal

November 7, 2026

To AC

Dear Area Chair,

We sincerely appreciate the valuable feedback from all reviewers and have addressed their comments in our rebuttal. However, we would like to bring to your attention a concern regarding Reviewer Ju5y's familiarity with our paper.

Reviewer Ju5y explicitly mentioned multiple times that the Reviewer is "completely unfamiliar with this problem setting" and therefore "cannot judge..." This suggests that the Reviewer's evaluation may not be well-informed. Moreover, the Reviewer's two main concerns were already discussed in the main paper, with clear references to the detailed discussion in the Appendix, yet the Reviewer did not seem to recognize these explanations.

Given this, we believe that Reviewer Ju5y's assessment may not be a reliable basis for evaluating our submission. We appreciate your consideration of this matter and thank you for your time and effort in handling our paper.

Reviewer 6m6G (rating 1)

We sincerely appreciate your valuable comments and suggestions!

Regarding the limitations of standard online RL algorithms in our framework: The core reason standard online RL algorithms cannot be directly applied is that **the online strategic interaction model under the source agent distribution (denoted by $\mathcal{M}^*(\mathcal{P}^s)$) differs from the one under the target agent distribution (denoted by $\mathcal{M}^*(\mathcal{P}^t)$)**. Simply ignoring feedback will lead to model misspecification and linear regret, as the existence of confounders leaves you a wrong distribution. Below, we provide a detailed explanation of this distinction and why it arises.

The knowledge transportability framework in this paper requires the principal to explore the optimal policy of an online strategic interaction model when interacting with a different agent type distribution than the one used for online data collection. Since the model under the source agent distribution $\mathcal{M}^*(\mathcal{P}^s)$ (which can also be viewed as an MDP per Section 3.1 of the paper) **is not the same as** the model under the target agent distribution $\mathcal{M}^*(\mathcal{P}^t)$, standard online RL algorithms cannot be used to find the optimal policy for $\mathcal{M}^*(\mathcal{P}^t)$.

A key motivation for studying knowledge transportability is that the agent population with which the principal interacts may change over time. For example, the personal attributes of job applicants at a company can vary across different periods. To capture such scenarios, our model allows the agent type distribution in the online setting to differ from that in the target setting. We also provide Example 3.2 in the paper as a motivating illustration.

If the source and target agent populations were **identical**, standard online RL algorithms could be directly applied. However, when the populations differ, solving the problem becomes considerably more challenging.

Regarding the experiments:

We conducted a small-scale experiment in the tabular setting, motivated by Example 3.1. Consider a company (the principal) is recruiting project managers (PMs, the agents), and the company needs to determine the salary of the PMs.

A corresponding bandit setting is constructed as follows: There is a single state and $H = 1$, with two candidate actions, H (high salary) and L (low salary). The agent's private type t can be either diligent (d) or lazy (l). The

principal's feedback, representing project performance, can be good (G) or bad (B). The reward function $R^*(a, e)$ takes as input the action a and feedback e. The feedback distribution $F(a, t)$ is defined as follows: - If $a = H$ and $t = d$, the feedback is always G. - If $a = L$ and $t = l$, the feedback is always B. - Otherwise, feedback is equally likely to be G or B.

In our experiment, we set $R^*(H, G) = 1.5, R^*(H, B) = 0, R^*(L, G) = 2, R^*(L, B) = 1$. The source agent distribution is 0.4 for d and 0.6 for l, while the target distribution is 0.8 for d and 0.2 for l. The principal's optimal action is H for the target distribution and L for the source distribution.

The empirical risk function $\hat{L}^k$ in episode k has the closed-form:

$$\hat{L}^k(R) = \max_{f=(f_H, f_L)} \sum_{\tau=1}^{k} \left(f(a_\tau)(R(a_\tau, e_\tau) - r_\tau) - \frac{f^2(a_\tau)}{2} \right).$$

Taking f_H as an example, the closed-form solution is $\sum_{a_\tau = H} R(a_\tau, e_\tau) - r_\tau$. The solution for f_L follows similarly. To simplify, we discretize the entries of $R(a, e)$ so that each takes values from {0, 0.5, 1, 1.5, 2}, leading to $5^4 = 625$ candidate models initially.

The following table summarizes the results, illustrating the convergence of our algorithm. We ran experiments with three random seeds, using $\beta = 600$ for 1800 episodes. In all cases, R^* was successfully recovered. The reported values reflect all three seeds.

— Episode — 300 — 600 — 900 — 1200 — 1500 — 1800 —

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— Average remaining model — 25,32,37 — 5,14,14 — 1,5,2 — 1,2,2 — 1,2,1 — 1,1,1 — — Average true action H taken (percentage) — 89,94,82 — 89,94,133 —389,151,313 — 689,451,613 — 989,751,913 — 1289,955,1213 — [Rui: maybe it comes from the difference between latex and markdown, this table is messy here.]

Further response We thank the reviewer for re-evaluating the paper and providing useful suggestions on the presentation of the formulation and experiments of the paper.

Regarding why assuming the principal know $\mathcal{P}^t$ but do not know $\mathcal{P}^s$: We address this by discussing both the underlying motivation (reflecting typical real-world scenarios) and the technical analysis.

In terms of the motivation, the discrepancy between $\mathcal{P}^s$ and $\mathcal{P}^t$ mirrors common economic events. It is crucial that the principal has some preliminary observations of the target population (e.g., through early surveys), as the quality of these observations greatly influences the effectiveness of the principal's policy. Without any knowledge of the target population, finding an optimal policy would be impossible. However, actively interacting with the target population is often challenging. For instance, a company may aim to recruit employees from a specific group but might only receive applications from the broader society, and our model is able to solve this problem as long as the target population can be approximated. [Rui: "making..." I'm a bit confised the meaning of this sentence. Does "filter and identify" mean solving MDP under the target distribution directly? If so, I think we should mention why our model can solve it.] Similarly, consider a scenario where a new medical treatment is tested in Country A, yet the government is interested in its effects in Country B; conducting the treatment in Country B may be impractical but obtaining demographics can be easier. [Rui: I guess it's better to say countries. And mention it's easy to get demographics but hard to conduct experiments] We have also provided some discussions in the Introduction [Rui: Introduction] and Appendix C.2 of the paper.

From a technical standpoint, the principal's optimal policy is inherently dependent on the target distribution, $\mathcal{P}^t$. Consequently, possessing knowledge of $\mathcal{P}^t$—or at least a reasonable approximation—is essential to determining the optimal policy. This assumption is standard in related fields, such as Myerson's auction theory and the coordination theory of principal-agent problems [1, 2].

[1]. Myerson, Roger B. "Optimal auction design." Mathematics of operations research 6.1 (1981): 58-73.

[2]. Myerson, Roger B. "Optimal coordination mechanisms in generalized principal–agent problems." Journal of mathematical economics 10.1 (1982): 67-81.

[Rui: End with "thank you"?]

Reviewer Ju5y (rating 2)

We sincerely appreciate your valuable comments and suggestions!

Regarding the difficulty in following the paper: Thank you for your feedback. We will address this issue in our revised version to improve readability and clarity. [Rui: I don't think we should say this. It sounds like we admit our paper is unclear.. maybe just say we will add a roadmap to help people unfamiliar with the literature to learn the logic.. BTW, I think it's important to leave AC a message, e.g., The reviewer mentioned that, "I am completely unfamiliar with this problem setting" and seems to have some misunderstandings on modeling. We hope the ACs/PCs will take this factor into account during the final decisions.]

Regarding the technical novelty compared to [1]: In contrast to Yu et al. [1], this work studies the **online** strategic interaction model. Below, we briefly highlight the key technical novelties, with further details provided in Appendix B.3.

1. To better align with real-world scenarios, our online model explicitly incorporates knowledge transportability. A key motivation for this is that the agent population with which the principal interacts may evolve over time. For example, the personal attributes of job applicants for a company may vary across different periods. Thus, our model allows the agent type distribution in the online setting to differ from the target agent type distribution of the principal.

2. The online strategic interaction model **cannot be directly solved using standard online learning tools** due to endogenous noise (i.e., confounding variables) and the knowledge transportability challenge. This work introduces novel techniques to address these issues:

- While both this work and [1] leverage the NPIV model for estimating the empirical risk function $\hat{L}$, we construct a distinct confidence set with a different concentration analysis. Specifically, [1] derives a concentration bound under an i.i.d. dataset assumption, whereas our work handles non-i.i.d. data. Furthermore, [1] constructs the confidence set by bounding the value difference between a candidate model and the minimizer of $\hat{L}$, whereas we construct our confidence set solely by bounding the value of $\hat{L}$ for candidate models.

- We establish a key error propagation technique from the NPIV estimator's error to the online regret using martingale concentration analysis with Freedman's inequality. Furthermore, we have developed cleaner proofs that rely only on the realizability assumption and the boundedness of zeros of a concave quadratic function, thereby simplifying the complex proof techniques in [1] that were based on the symmetric and star-shaped assumption.

Regarding the transferability assumption: Knowledge transportability is considered for two main reasons. First, in many real-world applications, the principal interacts with a diverse or evolving agent population, making this assumption highly practical. Our algorithm can also accommodate scenarios where the source population $\mathcal{P}^s$ changes across different episodes. Second, if the source agent population were identical to the target agent population, standard online RL algorithms could be directly applied. However, when these populations differ, solving the problem becomes significantly more challenging.

Regarding the "worst-case density ratio over all policies π and distributions ν" we note that this assumption can be generalized to an upper bound on the ratio of occupancy measures $d_h^{\pi,s}/d_h^{\pi,t}$ for any policy π. Equivalently, it corresponds to bounding the density ratio of the agent type distribution, $\prod_{i\le h}\mathcal{P}_i^s/\prod_{i\le h}\mathcal{P}_i^t$. Ensuring this ratio remains bounded is crucial for addressing distributional shifts between the source and target populations. Our problem is analogous to the covariate shift setting in unsupervised domain adaptation, where the model uses (s_h, a_h, e_h) to predict (r_h, s_{h+1}) based on underlying functions R_h^* and P_h^*. The source distribution is $d_h^{\pi,s}$, and the target distribution is $d_h^{\pi,t}$ for any policy π. A well-established condition for successful domain adaptation [2,3] requires a lower-bounded weight ratio, defined as $\min_X d_h^{\pi,s}(X)/d_h^{\pi,t}(X)$ for any measurable subset X of the input space. This precisely corresponds to the distribution shift term C_h^f in our paper, up to constant multipliers. Intuitively, without such conditions, knowledge transfer to the target domain would be infeasible.

[1]. Yu, Mengxin, Zhuoran Yang, and Jianqing Fan. "Strategic decision-making in the presence of information asymmetry: Provably efficient rl with algorithmic instruments." arXiv preprint arXiv:2208.11040 (2022).

[2]. Ben-David, Shai, and Ruth Urner. "On the hardness of domain adaptation and the utility of unlabeled target samples." International Conference on Algorithmic Learning Theory. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012.

[3]. Ben-David, Shai, and Ruth Urner. "Domain adaptation–can quantity compensate for quality?." Annals of Mathematics and Artificial Intelligence 70 (2014): 185-202.

Further Response:

Dear Area Chair,

We would like to call your attention again regarding Reviewer Ju5y's familiarity with our paper.

We have responded the concerns of the Reviewer in detail, and provided necessary references in the paper. The Review mentioned that the Reviewer is "completely unfamiliar with this problem setting" and therefore "cannot judge" the paper. Moreover, the Reviewer did not repsond to our responses.

Given this, we believe that Reviewer Ju5y's assessment may not be a reliable basis for evaluating our submission. We appreciate your consideration of this matter and thank you for your time and effort in handling our paper.

Reviewer o5zg (rating 3)

We appreciate your insightful comments and feedback!

Regarding the Markov policy class: We focus on the Markov policy class because any online strategic interaction model has an **optimal Markov policy**. This follows from the fact that an online strategic interaction model is equivalent to an MDP when the agent's private type distribution is given. Please refer to Section 3.1 of the paper for further details.

Regarding the comparison with [1]: While [1] also investigates a multi-agent generalization of the principal-agent problem, our work differs in formulation, methodology, and analysis.

- **Formulation**: [1] adheres to the traditional principal-agent framework, where the principal designs an incentive-compatible signaling scheme, and the agent responds based on the state and the principal's generated signal. In their setting, the principal has an information advantage over the agent, and their rewards depend on the global state and the agent's observable action. In contrast, our model fundamentally differs: the agent possesses private information hidden from the principal, and the agent's unobservable private actions directly influence the principal's reward. Additionally, both the principal and agents independently maximize their own utility functions without the incentive-compatible constraint. As a result, the decision-making processes in our model and theirs follow different logical structures.

- **Methodology and Analysis**: We propose a provably sample-efficient algorithm leveraging the NPIV method and optimistic planning. Our work includes a rigorous statistical analysis establishing the sample complexity of our algorithm. In contrast, [1] adopts a policy gradient approach and provides extensive empirical evidence to demonstrate its effectiveness.

Regarding the presentation: Thank you for your suggestions. We will revise the paper accordingly to improve clarity and readability. [Rui: same. I still disagree with mentioning clarity explicitly. Maybe just thank the reviewer for pointing out the typos and advice on writing and say we will fix them (or do anything else) in the camera-ready version.]

[1]. Lin, Yue, et al. "Information design in multi-agent reinforcement learning." Advances in Neural Information Processing Systems 36 (2023): 25584-25597.

Reviewer Whc9 (rating 3)

Thanks for your useful comments and suggestions!

Regarding the minimax estimators for IV:

We appreciate the suggestion to introduce a general framework rather than focusing solely on an exploration algorithm. We employ F-R duality to formulate a minimax estimator instead of using DeepIV or DFIV for the following reasons:

1. Our primary objective is to design **provably sample-efficient exploration algorithms** for real-world multi-agent decision-making systems with information asymmetry and knowledge transportability. The minimax NPIV method serves as a tool to address confounding issues in our framework. Additionally, analyzing its statistical

efficiency is crucial for proving the sample complexity of our exploration algorithm. While the minimax approach aligns well with our analysis, DeepIV and DFIV rely on deep neural networks, making statistical analysis challenging.

2. DeepIV and DFIV follow a 2SLS framework for IV regression in linear settings. Our framework generalizes beyond the linear case, as demonstrated in Section 5.2, where we show how our setting reduces to the linear case. Consequently, a more general algorithmic framework is necessary.

Regarding the bounded Lagrangian multipliers: The Lagrangian multipliers are bounded because the **concentration analysis of the NPIV model does not require them to take optimal values**. We discuss the relevant bounds in Lemma H.4 (Appendix H), where we show that they remain within a constant range.

Regarding the zero mean of the instrumental variable: We will expand Example 3.1 to illustrate a specific scenario where the zero mean assumption holds. Consider a company recruiting project managers (PMs) to lead projects. The company determines PM salary levels, with states corresponding to the company's status (e.g., stock price) and actions representing salary decisions by the board. The PMs' private types can be diligent or lazy. The zero mean assumption requires the company's reward to be approximated by an underlying reward function R^* under any agent type distribution. This is reasonable because the feedback e_h is able to capture the bias introduced by the agent type distribution. For example, when there are more diligent PMs, the company receives more positive feedback, leading to higher rewards with the same R^* but a higher probability of positive feedback. Generally speaking, we can "move" the mean into the reward R^* through the agent type distribution. Notably, the zero mean assumption does not require the noise to be zero mean given a specific agent type.

Regarding the experiments:

We conducted a small-scale experiment in the tabular setting, motivated by Example 3.1.

A corresponding bandit setting is constructed as follows: There is a single state and $H = 1$, with two candidate actions, H (high salary) and L (low salary). The agent's private type t can be either diligent (d) or lazy (l). The principal's feedback, representing project performance, can be good (G) or bad (B). The reward function $R^*(a, e)$ takes as input the action a and feedback e. The feedback distribution $F(a, t)$ is defined as follows: - If $a = H$ and $t = d$, the feedback is always G. - If $a = L$ and $t = l$, the feedback is always B. - Otherwise, feedback is equally likely to be G or B.

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Regarding the presentations and references: Thank you for your valuable suggestions. We will revise the presentation and incorporate the recommended references accordingly.